Task 4 – Common Network Security Threats

Cyber Security Internship

Prepared as an internship task submission

1. Introduction

Network security is not only about protecting a computer from a virus. A network can be affected by many different threats, including malicious software, phishing, password attacks, denial-of-service attacks and weaknesses in applications. In this report I looked at common threats and the basic measures that can reduce their impact.

2. Malware

Malware is software created to damage systems, steal information or perform unwanted actions. Common examples include viruses, worms, trojans, spyware and ransomware. Keeping software updated, using endpoint protection and avoiding untrusted downloads are practical ways to reduce the risk.

3. Phishing

Phishing uses fake messages, websites or emails to trick people into giving away information. The attack often depends on urgency or fear. Checking the sender, inspecting links carefully and using multi-factor authentication can make these attacks harder to succeed.

4. DoS and DDoS

Denial-of-service attacks try to make a service unavailable by overwhelming it with traffic or requests. DDoS attacks can come from many systems at once. Filtering, rate limiting, resilient infrastructure and dedicated DDoS protection can help reduce the impact.

5. Man-in-the-Middle Attacks

In a man-in-the-middle attack, an attacker attempts to intercept communication between two parties. Using HTTPS, secure networks and properly configured encryption helps protect communications.

6. Password Attacks

Weak or reused passwords can be attacked through brute force, dictionary attacks or credential stuffing. Strong unique passwords, password managers, account protections and multi-factor authentication are useful defenses.

7. Application-Based Threats

SQL injection and cross-site scripting are examples of attacks against web applications. Input validation, parameterized queries, output encoding and secure development practices help reduce these risks.

8. Insider Threats

A security incident does not always come from outside an organization. An employee or other authorized user can accidentally or deliberately expose information. Least privilege, monitoring and security awareness are important controls.

9. Zero-Day Vulnerabilities

A zero-day vulnerability is a previously unknown or unpatched weakness that can be exploited before an effective fix is widely available. Organizations can reduce exposure through monitoring, layered defenses, rapid patching and vulnerability management.

10. Prevention and Best Practices

• Keep operating systems and applications updated.

• Use strong, unique passwords and multi-factor authentication.

• Back up important information regularly.

• Use firewalls and endpoint protection.

• Train users to recognize suspicious messages.

• Monitor systems and investigate unusual activity.

• Use encryption for sensitive information.

11. Personal Takeaway

The biggest lesson from this report is that network security needs several layers. A firewall alone cannot stop phishing, and antivirus alone cannot solve weak passwords. Technical controls and user awareness need to work together.

12. Conclusion

Common network threats can affect confidentiality, integrity and availability. Understanding how these attacks work makes it easier to choose appropriate controls and respond to incidents before they become larger problems.